

2026.05.07

PROJECT PROPOSAL

Digital Multi-FX Audio Processor

Configurable Real-Time DSP Device

Presented by Team 5

CONTENTS

01. Introduction

02. WHY - Motivations

03. WHY - FPGA vs MCU

04. Objectives

05. HOW - Framework

06. HOW - Architecture

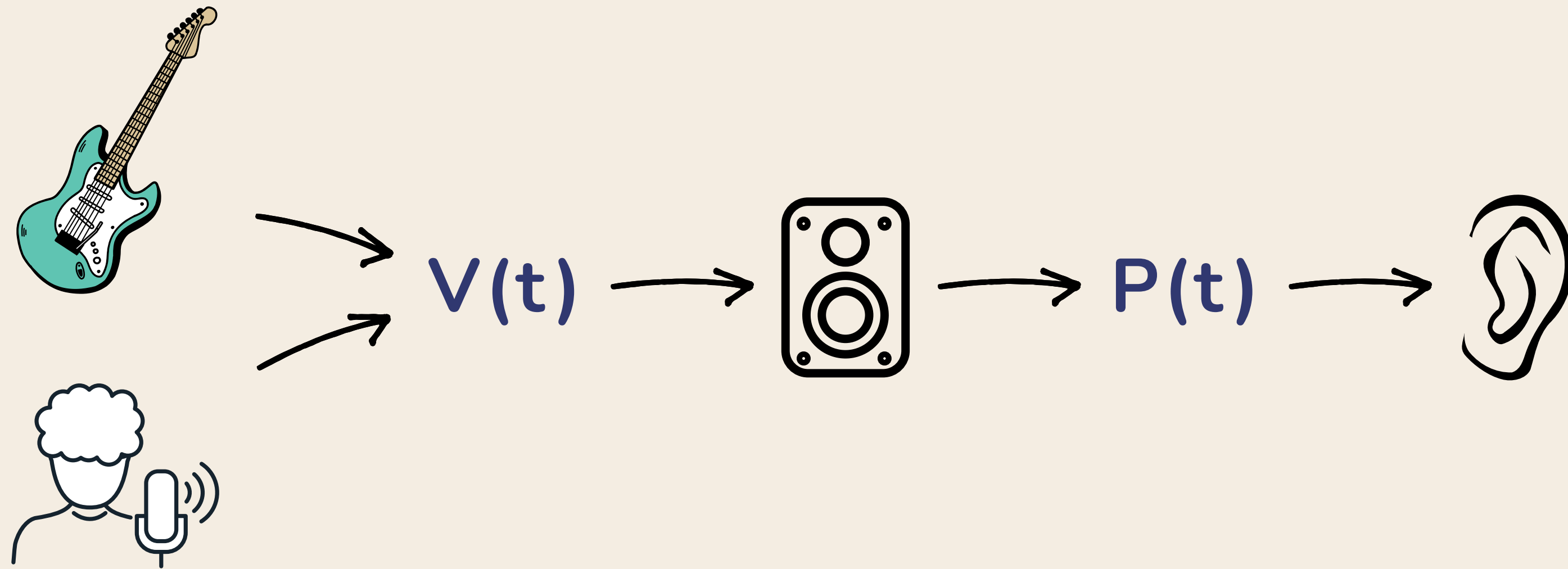
07. Roadmap

08. WHAT - Problems

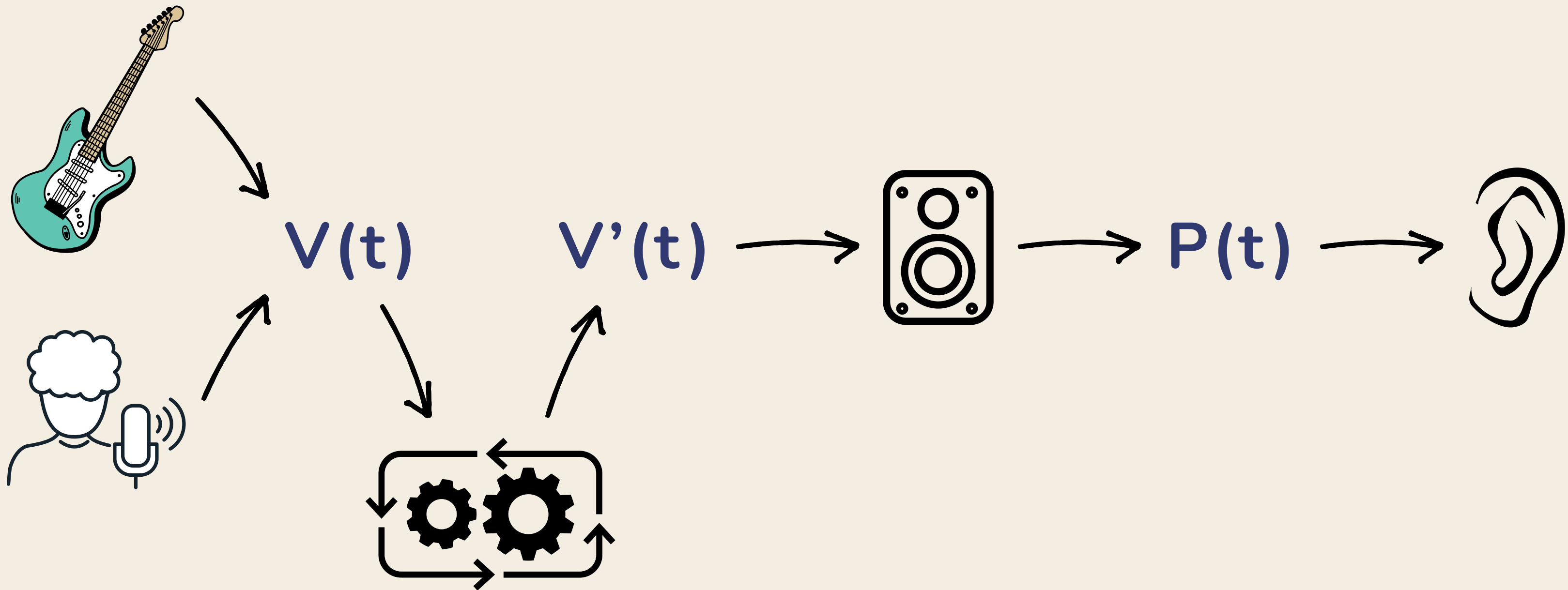
09. WHAT - Hardware

10. WHAT - Budget Plan

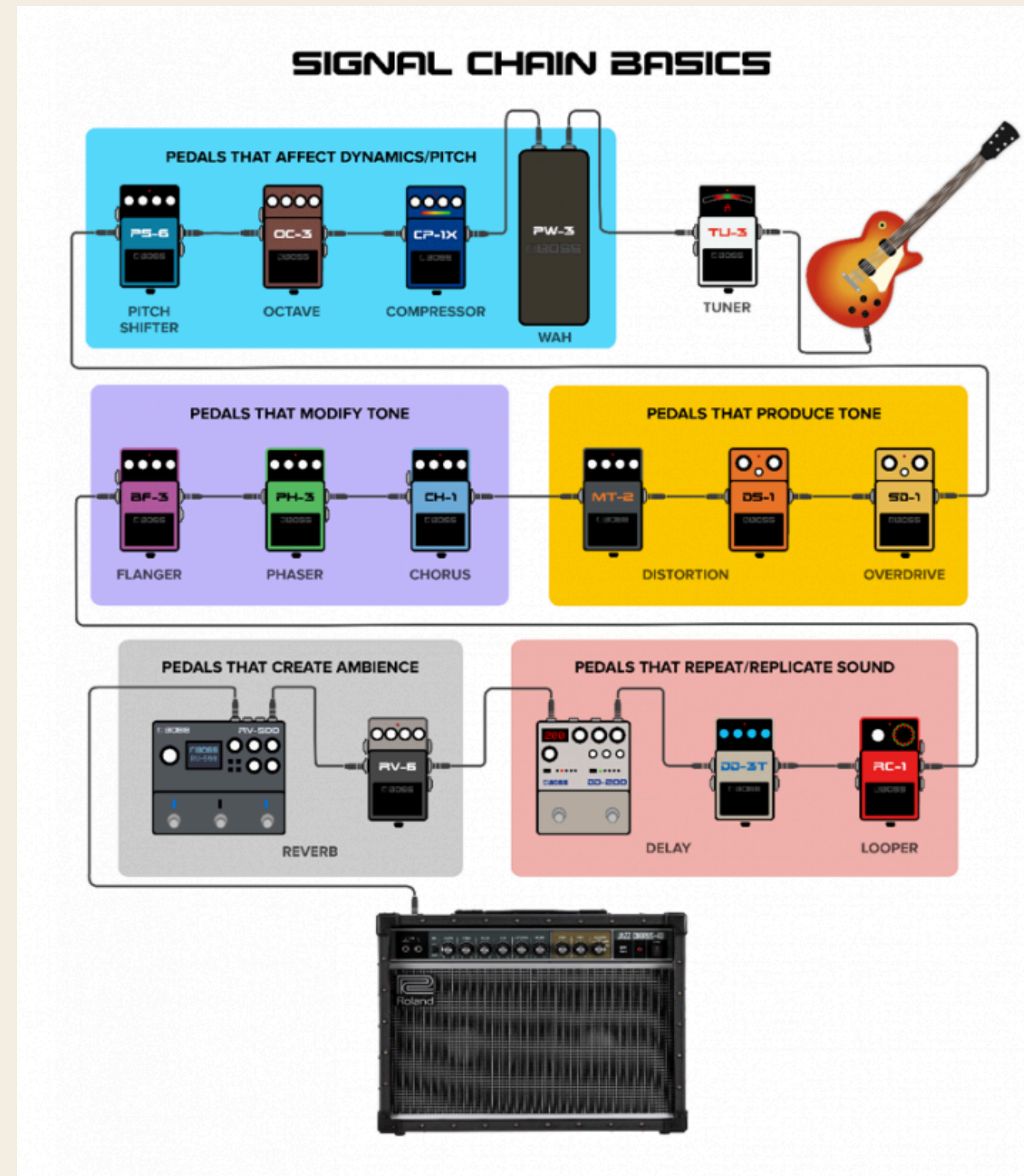
INTRODUCTION



INTRODUCTION



INTRODUCTION



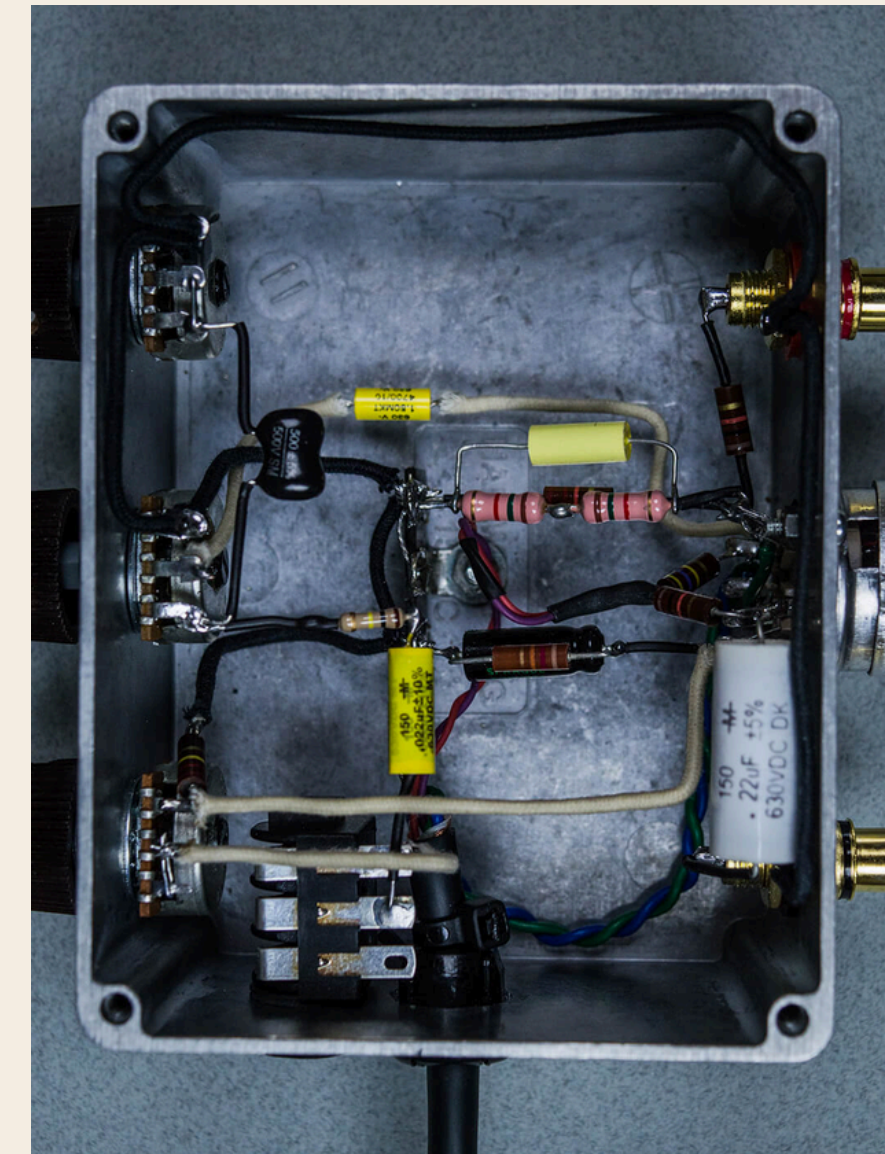
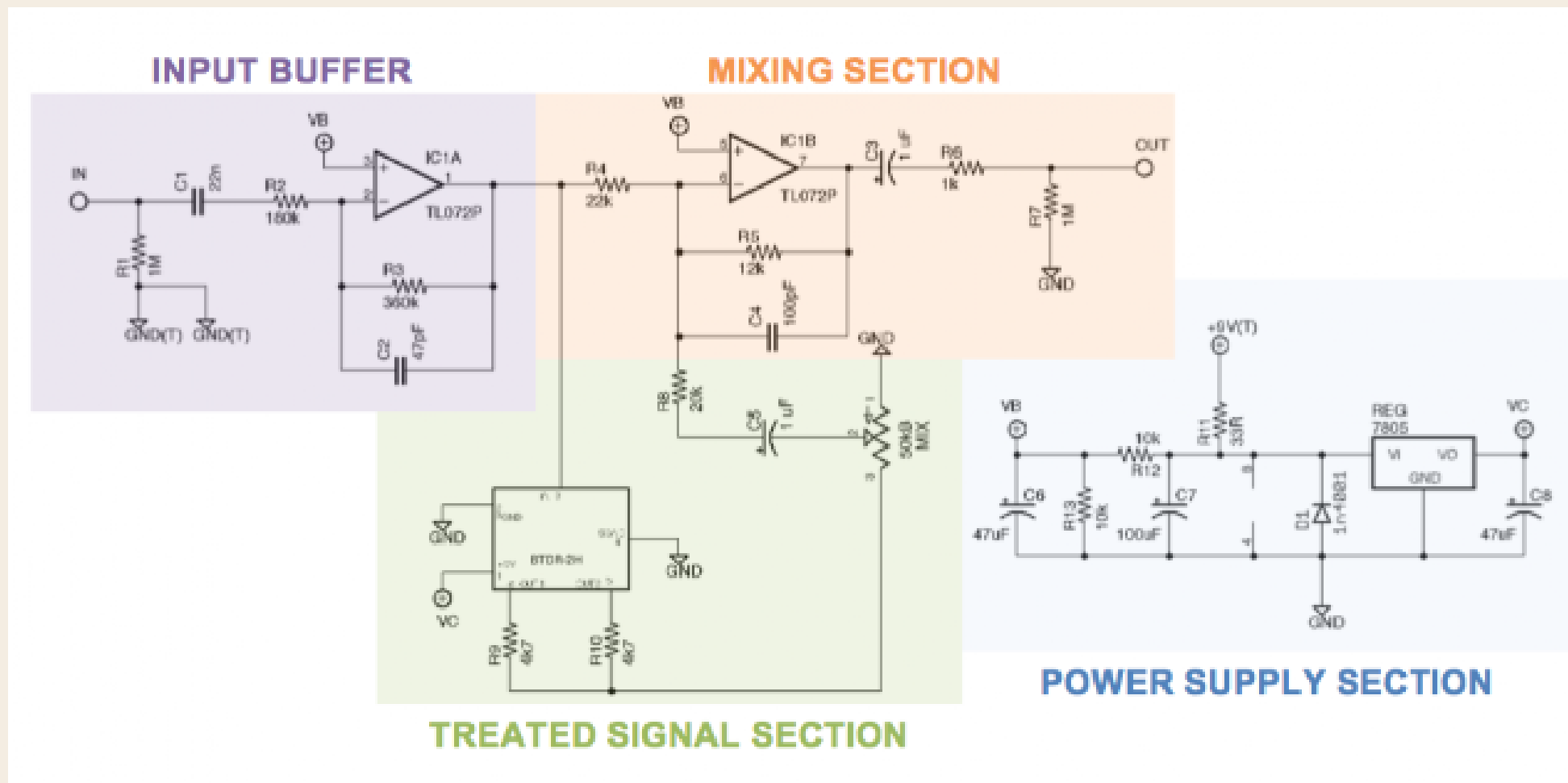
INTRODUCTION

Pedal Rack



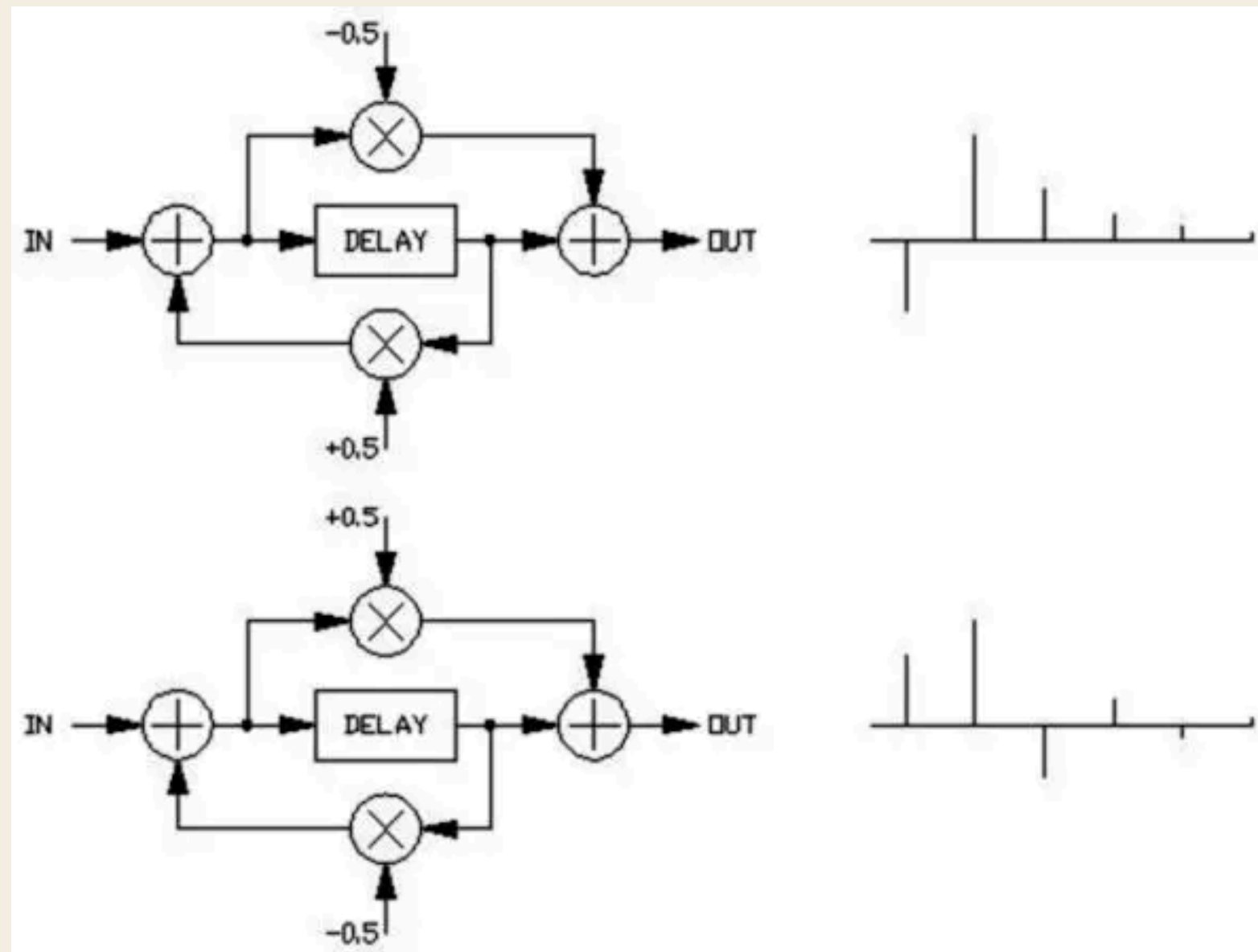
INTRODUCTION

Analog Approach: Reverb



INTRODUCTION

Digital Approach: Reverb



INTRODUCTION

Digital Multi-FX Pedals (綜合效果器)



Pros: 一顆抵多顆、方便攜帶、調整自由度高

WHY – MOTIVATIONS

1. We can utilize Lab3 as a head start
2. Great chance to learn about audio signal processing with DSP
3. A Project better on FPGA compared to MCU (Arduino, ESP...)

WHY – FPGA VS MCU

Biggest problem of most
audio DSP plugins / pedals: **Latency**

FPGA allows **parallel computation**

→ we can **pipeline** the signals between blocks

→ significant latency improvement (in theory)

OBJECTIVES

Input: Guitar / Mic

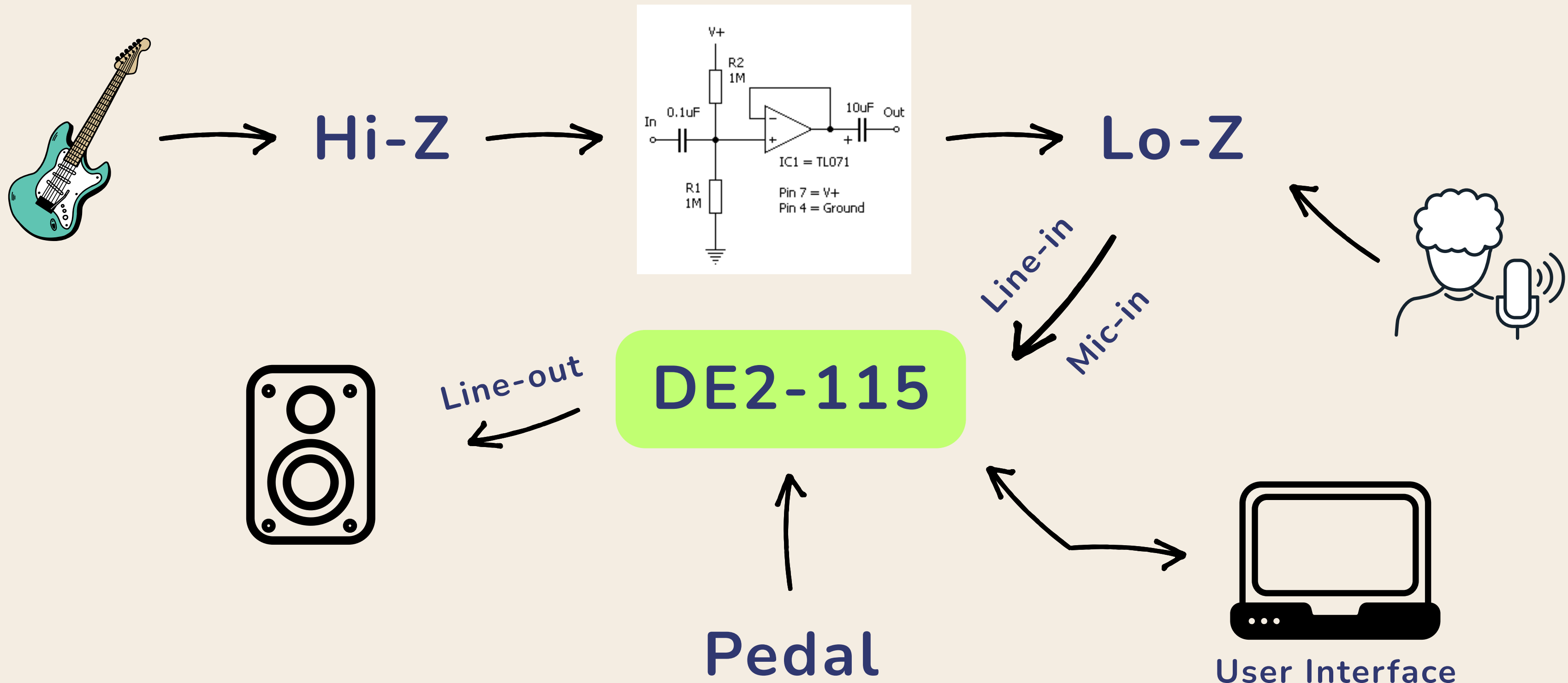
Output: Speaker / Headset / Amp

Objective 1: Implement several **common audio EFXs as individual modules**, which can be chained in arbitrary order.

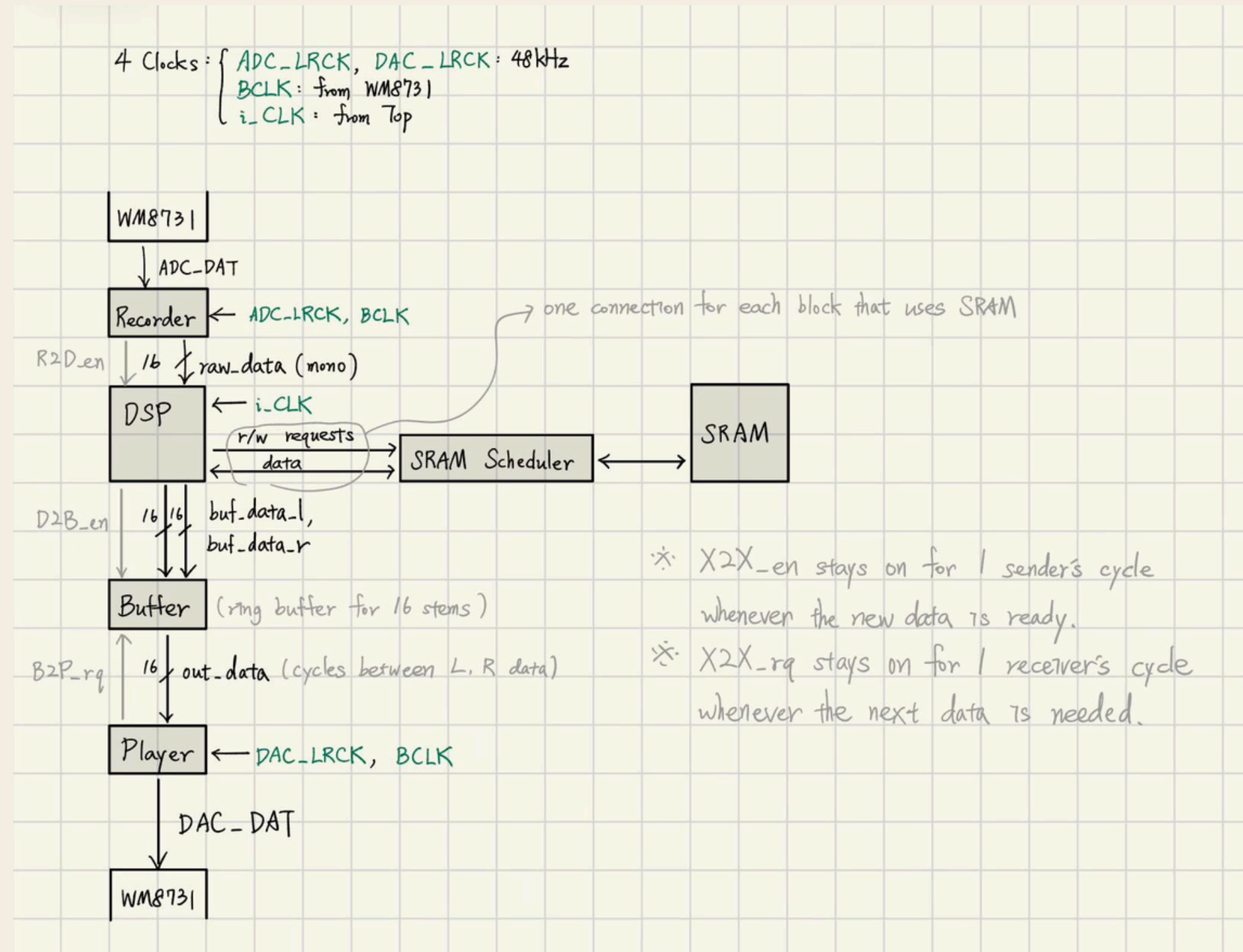
Objective 2: Implement a **computer UI** for real-time parameter adjustment.

Objective 3: Allow user to save parameters / chaining order as presets, and conveniently **switch between presets with a pedal**.

HOW - FRAMEWORK



HOW - ARCHITECTURE



HOW - ARCHITECTURE

Block Type		
SRAM Access $I \rightarrow 0$ mono $\rightarrow$ mono mono $\rightarrow$ stereo	$\times$ 優先 先做到硬體再說	$\checkmark$ 其次

ROADMAP

I - Digital EFX

clean tone, volume
overdrive, distortion, fuzz
delay, reverb
EQ, chorus, flanger, phaser

II - Module Chaining

Implement staging unit

III - Computer User Interface

Real-time parameter configuration
Preset saving / loading

IV - Pedals

Preset switching pedal
WAH - Pedal / Volume Pedal

WHAT - PROBLEMS

Problems yet to be solved:

1. SFFT on FPGA
2. Connection with computer / UI Design
3. Hardware Design

WHAT - HARDWARE & BUDGET PLAN

Input Buffer	< 300 NT
Pedals	< 1000 NT
Decorations	< 300 NT
Total	< 1600 NT

WHAT – BONUS

- Tuner
- Chord Identifying
- Tone Replacement

Disclaimer: All items mentioned here are additional ideas for the project, which will NOT be pursued or attempted if we find ourselves too busy with executing the original plan. This part is NOT planned at all.

2026.05.07

THANK YOU

Thanks For Listening

Presented by Team 5